Power of Prevention

Diabetes and Heart Health

Diabetes and its complications are preventable.

Heart attacks, angioplasty and bypass surgeries
Are preventable. To a major extent.

The answer is in your hands, ... And
Samatvam – Jnana Sanjeevini is there to help you.



Samatvam

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  E-mail : samatvam@gmail.com
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DIABETES
and
HEART HEALTH

The answer is in your hands



Since 29 Years
1987

Samatvam - Jnana Sanjeevini
Bengaluru - India

Here we are

Samatvam
Endocrinology Diabetes Center

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Diabetes
- ► Day Care Unit
- ► Annual Health Review
- ► Heart-Kidneys-Eyes-Feet
- ► Family Education Unit

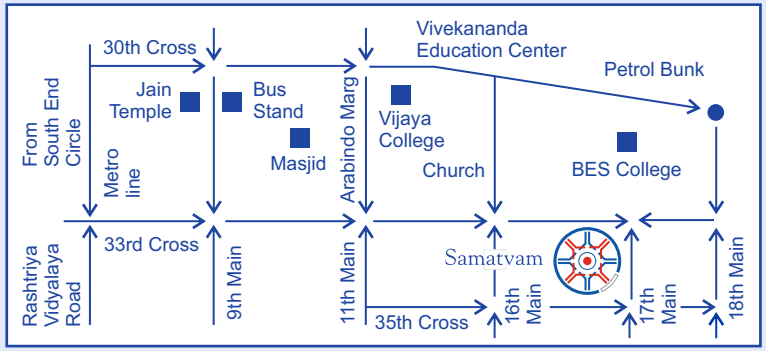
Endocrinology
- ► Hormone Laboratory
- ► Thyroid-Obesity-Lipids
- ► Infertility-Reproduction
- ► Growth-Development

Websites: samatvamdiabetes.com | jsindia.org | dishastars.org

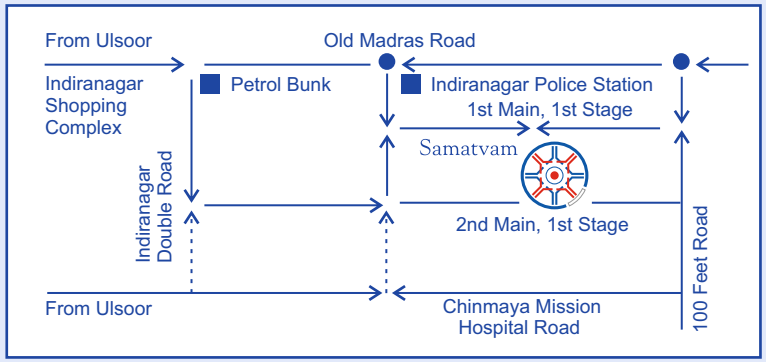
24 Hour Diabetes Help Lines:
9449994978, 9448356575

How to get here

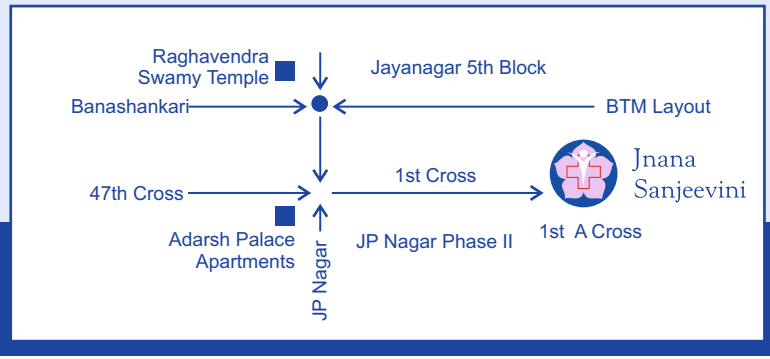
Samatvam
Jayanagar



Samatvam
Indiranagar



Jnana Sanjeevini
JP Nagar



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Dr. Tejaswini Patil	MD, FEDM, FACE
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Dr. Pushpa Ravikumar	MBBS, FEDM, FACE
Dr. Lakshmi V Reddy	MBBS, FEDM, FACE, (Fellow ISPAD)
Dr. K M Chandrika	MBBS, FEDM, FACE, (Fellow ISPAD)
Dr. K Sumathi	MBBS, FEDM, FACE
Dr. S Chandraprabha	MBBS, FEDM, (FACE)
Dr. Vasanthi Nath	MBBS, DGO, FEDM, (FACE)
Dr. B V Reshma	MBBS, FEDM, (FACE)
Dr. Babitha Devi	MBBS, FEDM, (FACE)
Dr. Vasudha Srikanth	MBBS, FEDM, (FACE)
Dr. A Sumithra	MBBS, FEDM, (FACE)
Dr. M D Chitra	MBBS, FEDM, (FACE)
Dr. Kavitha muniraj	MBBS, FEDM, (FACE)
Dr. Amit Kumar Sinha	MBBS, FEDM, (FACE)
Dr. M Sujatha	MBBS, FEDM, CDE
Dr. Vasanthakumari	MBBS, FEDM, CDE
Dr. K V Rajiv	MBBS, FEDM, (FACE), (Fellow ISPAD)
Dr. C S Muralidhara Krishna	MD, FEDM, (FACE), (Fellow ISPAD)
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Samatvam – Jnana Sanjeevini

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Ms. H K Vasanthalakshmi	B.Sc, CDE, PA
Ms. Nandini Jayaram	B.Sc, CDE, PA
Ms. Rajeshwari Ashok	B.Sc, DMLT, CDE, PA, (Award ICE)
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Ms. Nagamani Srinivas	BNYS, CDE, PA
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Ms. Rohini Manjunath	B.Sc, CDE, PA, (Award ISPAD)
Mr. D Mallikarjun Rao	PG, DFD, Podiatry
Ms. Sujatha Swamy	B.Sc, CDE, PA
Mr. Owais S M H	B.Sc, CDE, PA, (Award ICE)
Ms. L B Akshaya	B.A, DNHE, CDE, PA, (Award ICE)
Ms. B Marimuthu	PUC,DHA, CDE, PA, (GNM), (Award ICE)
Ms. Mary Priya	S.S.L.C, CDE, PA
Ms. S Varalakshmi	S.S.L.C, CDE, PA
Ms. M Suma	S.S.L.C, CDE, PA
Ms. Padmaja Srinivas	MA, LLB, CDE, PA

Our Achievements

Jnana Sanjeevini Diabetes Hospital is a dedicated public service project of Samatvam Trust, initiated in 2003, at JP Nagar, Bangalore. The aim is to provide multidisciplinary care for diabetes and related health problems 'all under one roof'. In 2012, Samatvam completed 25 years of public service, touching the health and lives of more than 100000 fellow citizens with diabetes - adults and children, poor and rich.

Our service accomplishments include, India's First / Earliest / Structured:

- Diabetes Patient Education Program (Hospital, Clinic and Community Based)
- Diabetes Educator/Counselor
- Childhood Diabetes Clinics and Camps – Residential
- Diabetes Support Groups – Patient Parent
- Diabetes Foot Nurse / Podiatrist
- 24 hour FREE Diabetes Helpline – World's longest and busiest
- Mass Media Health Education – Eg: Live interactive TV programs
- Insulin Pump Program

Integration of Science and Spirituality in the context of healthcare is our foundation.



Community Programs

"No one will leave **Jnana Sanjeevini** uncared and unprovided, irrespective of their socioeconomic status."

Project DISHA

Beneficiaries: Type I (insulin dependent) diabetes children and youth {3000}

- FREE Childhood Diabetes Clinic
- FREE Insulin lifeline: Insulin for every child
- Insulin PLUS Education

Project DOSTI

Beneficiaries: Type II (non – insulin dependent) diabetes adults {30000}

- FREE Adult Diabetes Clinic
- FREE medicines: Health for helpless

Project DEEPA

Beneficiaries: Diabetes physicians and health counselors – professionals {250}

- Observerships - Internships - Fellowships
- Continuing medical education - training

Diabetes Challenge

Diabetes is the biggest public health crisis of the 21st century and it continues to grow at epidemic proportions. India bears the highest burden of people with diabetes (60 million in year 2011, increasing to 100 million by year 2030) in the world. Nearly 25 -50% of the people having diabetes don't even know it. Diabetes increases the risk for a number of serious, even, life threatening complications, including heart disease, strokes, blindness, kidney failure, gangrene and leg amputations.

People with diabetes (compared with people without diabetes) are:

- 5 times more likely to suffer heart attacks (# 1 cause of death);
- 3 times more likely to suffer stroke;
- 30 times more likely to require leg amputation;
- 20 times more likely to develop kidney failure;
- 25 times more likely to develop blindness.

Diabetes shortens life span by a decade (on an average), and causes economic disability and death, in the prime of life.

Many children with diabetes in India die before they are diagnosed. Even those who are lucky enough to be diagnosed, face major social and economic hardships and discrimination (in education and employment).



Power Of Prevention

Diabetes complications and organ damage STEPS for PREVENTION

Preventive Steps	Heart	Brain	Feet	Kidneys	Eyes	Nerves
Tight blood glucose control	√	√	√	√	√	√
Tight blood pressure control	√	√	√	√	√	√
Normal blood cholesterol & fats	√	√	√	√	√	√
No smoking	√	√	√	√	√	√
Aspirin	√	√	√	√	√	√
Annual Health Review	√	√	√	√	√	√

Important to do

- Eat healthy foods in right amounts at set times.
- Follow your physical activity routine.
- Maintain your body weight at the ideal level.
- Take your medicines everyday. Take the right amount at the right time.
- Test your blood glucose regularly. Test more often when you are sick.
- Test your urine for ketones when your blood glucose is over 240 mg/dl.
- Keep your blood pressure and blood fats near normal.
- Check your feet and skin daily for injury and infection.
- Get eye and teeth exams every year.
- Do not smoke. Smoking is most dangerous to health and life.
- Annual Health Review is a must for everybody with Diabetes.

Keep learning and caring about your diabetes and its treatment.

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DIABETES : The answer is in your hands

Name :_____

Age(Yr.): _____ Sex: _____ Hosp. No. _____

Referring Physician : _____ Date : _____

"A person with Diabetes can live a full and productive life."

Health Education

The prime purpose of diabetes patient and family education is to make each patient as knowledgeable and self sufficient as possible, so as to:

A) Achieve best possible diabetes control (short term);

B) Reduce and/or postpone the various adverse effects or complications of diabetes (long term);

C) Empower patient and families to assume proper control over the day to day management of diabetes and overall health.

Health Education check list:

Topic	Initial/review (Dates)				Remarks
Achievement level	0	1	2	3	
1. Diabetes-An overview					
2. Meal plans Exchange lists Carbohydrate Counting					
3. Physical activity					
4. Diabetes pills					
5. Insulin injections Storage Mixing Injection technique Insulin pens Insulin pumps					
6. Monitoring diabetes control Urine tests Blood tests: Meter Hemoglobin A1c					

Topic	Initial/review (Dates)				Remarks
Achievement level	0	1	2	3	
7. Special problems Hypoglycemia Glucagon injections Ketoacidosis Ketones testing					
8. Sick day guidelines NEVER OMIT INSULIN					
9. Insulin adjustments Algorithms					
10. Personal hygiene					
11. Foot care/shoes					
12. Travel					
13. Diabetes and pregnancy					
14. Chronic complications					
15. Diabetes ID card					
16. Literature checklist Diabetes teaching guide					
17. Self care basic					
18. Self care intensive					
19. Blood pressure					
20. Cholesterol, fats					
21. Smoking					
22. Alcohol					
23. Heart health					
24 Stress management					
25. Yoga -Spirituality					
26.					
27.					

Achievement level : * Areas need instruction. 0 = guide given; 1 = initial teaching done; 2 = further teaching needed/reviewed; 3 = can adequately describe or perform.



Nature of Diabetes

- Diabetes is a condition in which either your pancreas does not produce enough insulin or it produces sufficient insulin but the cells of your body are unable to use it properly.
- Diabetes takes away your body's ability to keep the amount of glucose in the blood at a healthy level. When your blood glucose remains out of control you have a greater chance of developing certain health problems.

Symptoms of Diabetes

You may feel more thirsty than usual and pass more urine quite often. You may experience loss of weight, blurred vision, susceptibility to some infections, delay in wound healing, tiredness and sleepiness all the time. However, nearly half of people with diabetes, may experience no symptoms at all. Thus regular and periodic screening of blood glucose levels is very crucial for early detection of diabetes in the population.

Types of Diabetes

Insulin is a hormone produced by our pancreas. Insulin helps to convert sugar or glucose (basically food we eat) into energy. The energy is carried to the millions of cells in our body. Insulin is released into our blood stream whenever the glucose level goes up (eg, after a meal). As your blood glucose level drops, so does the secretion of insulin from your pancreas. As insulin circulates, it acts like a key by unlocking microscopic doors that allow glucose to enter your cells. Insulin lowers the amount of glucose in your bloodstream.

There are two main types of diabetes:

- Type I Diabetes is also called IDDM (Insulin Dependent Diabetes Mellitus). The body makes little or no insulin.

  If you have Type 1 diabetes, your pancreas produces very little, if any, insulin. This type occurs most often but not exclusively in children and young adults. People with this type of diabetes must and should inject themselves daily with insulin, without which their survival is not possible.
- In Type I Diabetes, you can have moderately high or severely high blood sugar with a serious condition known as ketoacidosis. This may be detected during the initial diagnosis, after infection or other severe stressful situations. Response to treatment with injection insulin is excellent and life saving.

  Type II Diabetes is also called NIDDM (Non Insulin Dependent Diabetes Mellitus). The body makes some insulin, but cannot use it well.

If you have Type 2 diabetes, your pancreas produces insulin but it is either not enough for the present needs of the body, or, if it is, the cells of your body are unable to use it properly. It is called non-insulin dependent because it can be usually controlled without the use of insulin injections.

- Most often high blood glucose level may be the only sign of diabetes. Some people have other conditions also, like high blood pressure, obesity and high blood lipids.

Complications of Diabetes

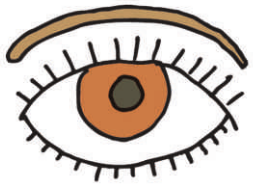
- People with diabetes are more likely than others to have certain health problems known as complications. These include: heart attacks, strokes, foot/leg removed by surgery (amputated); frequent infections; eye problems leading to blindness; kidney diseases.
- The above health problems are caused by several factors like, damage to blood vessels, damage to nerves, decrease in ability to fight infections etc.
- Aim at optimal diabetes control and avoiding/normalizing risk factors (like smoking, high BP, high blood fats, overweight, physical inactivity, alcoholism etc.) all of which will help reduce the chances of complications significantly.

Control of Diabetes

There is no cure for diabetes. The goal of treatment is to control it, which is to keep the blood glucose level near normal. The following table indicates the average target for blood glucose levels. However, the target can differ from person to person. Your doctor will help you in this and will provide tips to achieve the set target.

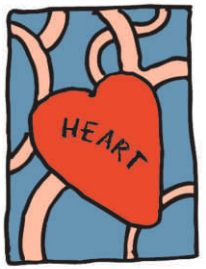
Blood glucose mg/dl	Fasting mg/dl	Two hours after meals	HbA1C %
Normal	Upto 100	Upto 140	< 6
Acceptable	Upto 126	Upto 200	6 - 8
Poor	Over 126	Over 200	> 8

Complications of Diabetes



eye problems – even blindness

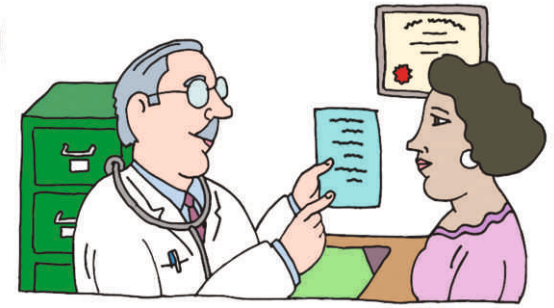
heart disease





foot problems – even losing a foot or leg

Control of Diabetes









- Besides working to control your blood glucose, there are other actions you must take to reduce your risk for health problems.
- Have your blood pressure and blood fats checked regularly. If they are high, find out what to do and stick to the treatment.
- Have a complete eye exam every year. This means more than just an eye glass check. The exam is done in a darkened room. Photos of the inside of the eye may be taken.
- Check your feet and skin everyday. If you are injured, have pain, or see signs of infection (redness, swelling or pus), get treatment right away..
- Do not smoke. Smoking and diabetes are a dangerous combination, especially if you also have high blood pressure. Stop smoking right away.
- Have your blood and urine checked each year for signs of kidney damage. Consult your doctor to know what the results mean and learn what you can do to protect your kidneys.
- Avoid alcohol. Alcohol affects your heart and also affects blood fat levels.

Treatment of Diabetes

There are several major treatment tools to control your diabetes. All the tools are interrelated and must be included in your treatment program. Medication alone will not provide control over diabetes.

- Education
- Meal Planning
- Physical Activity
- Diabetes Pills
- Insulin Injections
- Newer Diabetes Medicines
- Insulin Pumps
- Diabetes Related Tests

Treatment of Diabetes



Education

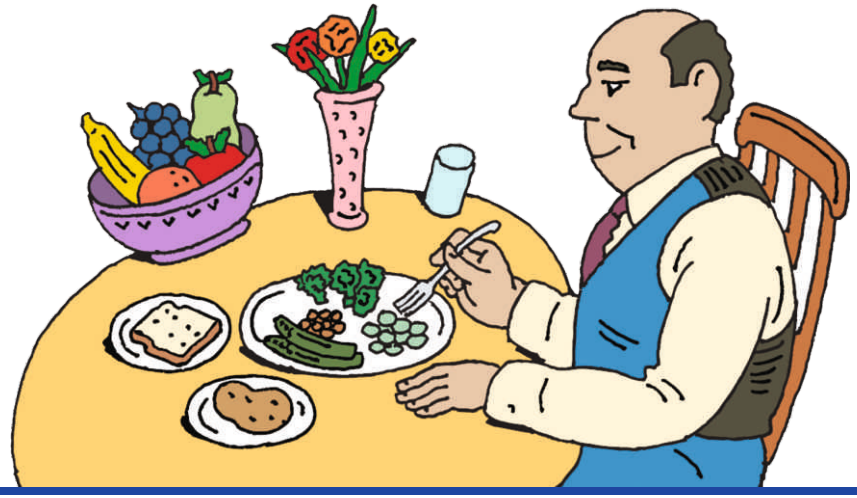
Education is the starting point. Learn to use other treatment tools. Learn about yourself; how diabetes affects you, your feelings, your family, and how to find help when you need.

Meal Planning

- Meal planning means choosing healthy foods, eating the right amount of food for you and eating your meals at certain set times. The same healthy diet that is best for everyone else is the best for controlling diabetes.
- Eating healthy foods is one of the most basic and important tools of diabetes management. The right food choices can help control blood glucose and protect long term health. Diabetes medicines must be balanced with the type, timing and amount of food eaten. An eating plan to lose excess weight and reduce salt intake is an important part of a program to control blood pressure. Eating less animal fat and avoiding high cholesterol foods help control blood fats.
- Eat a variety of foods every day. Choose from different cereals (rice, wheat, ragi), pulses and dhals, fruits, vegetables, low fat milk products, and low fat protein foods. Use a glucose substitute if necessary, instead of table sugar. Eat more fiber. Choose whole grain breads instead of white bread. Eat the edible skins and peels of fruits and vegetables. Limit foods prepared with large amounts of added salt like pickles, papads, canned soups and smoked meats.
- Eat meals and snacks at regular times to balance medicines and control hunger.

Meal Planning





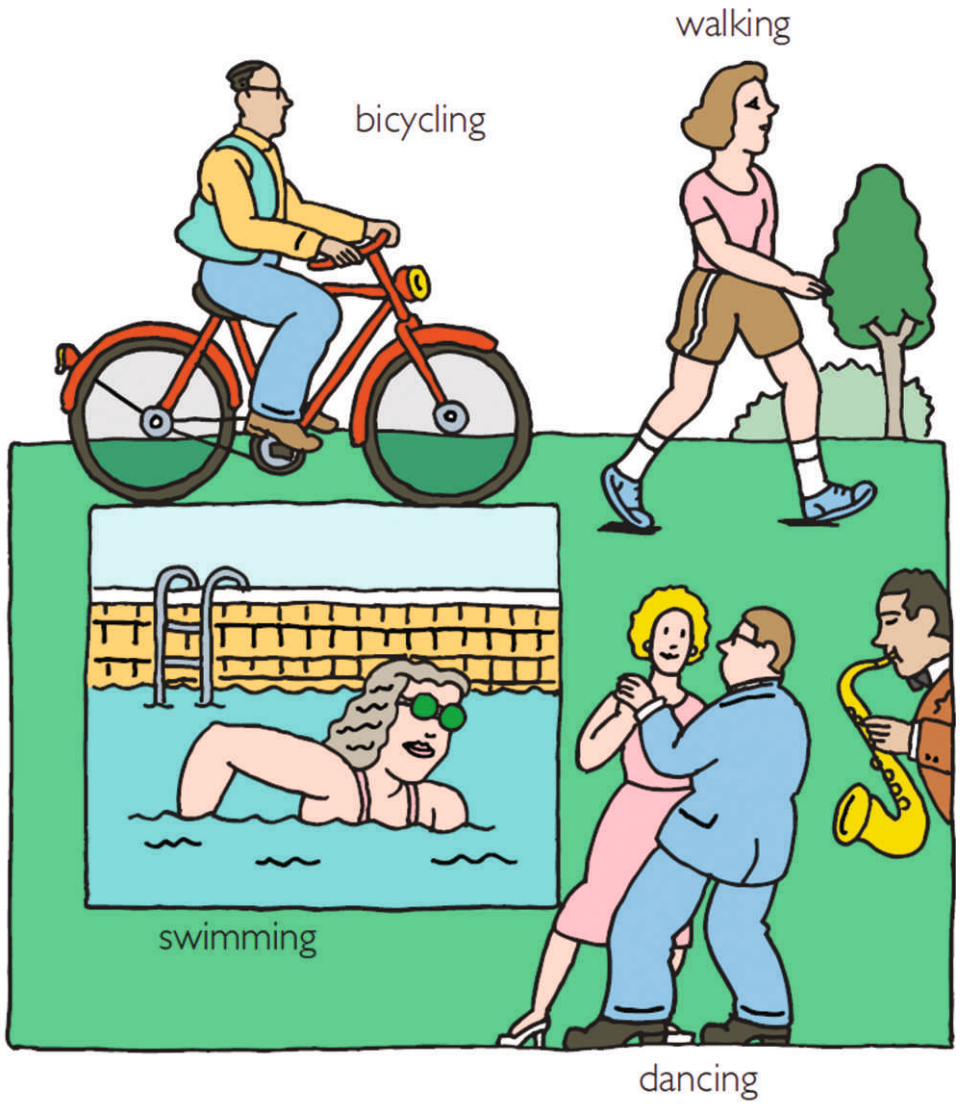
Physical Activity

- Physical activity can help you manage your overall health, and controls your diabetes. It can keep your weight under check and the blood glucose at levels in the normal range. It also helps you to cope better with stress.
- Exercise is important for maintaining your overall health as well as controlling your diabetes. Exercise preserves and increases your muscle tone and strength, provides exercise for your heart and blood vessels and increases your lung efficiency. For the person with diabetes, exercise has additional benefits. Exercise lowers the blood glucose levels. Exercise makes it possible for the cells of your body to use glucose more easily. Exercise can also lower the cholesterol and triglycerides in your blood. A regular exercise program can be a valuable addition to a weight reduction programme for people who are overweight.
- The best exercise for a person with diabetes is an activity which uses large amounts of energy over a shorter period of time, such as brisk walking, running, swimming, bicycling, tennis etc. These activities help maintain good blood glucose control.
- One can start the exercise program with 10-20 minute duration, which can be gradually increased to at least 60 minutes per day.
- The ancient Indian system of yoga has numerous health benefits (physical exercise, as well as spiritual - relaxation, stress management).

Diabetes Pills

- Diabetes medicines lower the level of sugar in the blood. Some people need insulin injections to replace or add to their body's own insulin. Other people use diabetes pills called oral hypoglycemic agents (OHA) which improve insulin secretion from your pancreas or work with body's insulin and make it more effective. Take them every-day in the right amount at the right time.
- They stimulate the pancreas of some people with diabetes to release additional insulin (sulfonylureas, newer gliptins). They also make the insulin more effective by helping the cells of the body use insulin properly (biguanides, sulfonylureas).
- They are useful in patients with non-insulin dependent diabetes. They must not be used in pregnant women and children. They are not insulin or even remotely related to it. There is no insulin which can be taken by mouth at the present time.
- Diabetes pills must not be used or should be used with extreme caution in people with kidney or liver problems. These tablets also should not be used during surgery or serious illness. Consult your physician.

Physical Activity



Take sulfonylurea group of tablets (eg: glibenclamide, glimepirede, glipizide) 20-30 minutes before meals. Biguanide group of tablets (eg: metformin) must be taken immediately after food.

- Be prepared for low glucose reactions even while taking diabetes pills. Never skip meals after having taken the tablets. Do not change the brands of tablets on your own. They are not interchangeable. Consult your physician.

Insulin Injections

- All people with Type 1 diabetes must inject themselves daily with insulin. In many people with type II diabetes, over a period of years, diabetes pills may become ineffective in controlling blood glucose levels adequately. Then you will need to take insulin injections in addition, to achieve the best possible control of blood glucose.
- Insulins were feverously obtained from the pancreas of pigs or cows. These are called the porcine or bovine insulins. Now, Human Insulin and analogues are being produced by genetic engineering.
- Insulin is measured in units just as milk is measured in liters. Insulin is produced in two different strengths. U-40 and U-100. U-40 is the most commonly used insulin strength in India. In addition to different strengths, there are different types of insulins called short acting (regular), intermediate acting (NPH, lente), long acting and newer analogues (designer insulins)
- Insulins when exposed to excessive heat or cold lose their potency. Store insulins carefully. You should take insulin injections generally about 20-30 minutes before food. Your doctor will give you the number of units you need to take, the type of insulins and the frequency of administration necessary to achieve adequate diabetes control.

Newer Diabetes Medications

- We have seen several new, exciting classes of drugs in the past decade. New drugs and types of insulin have made treatment better. In some cases, it is simpler, too.
- Gliptins [Incretin enhancers - DPP-4 inhibitors]: Some people with type 2 diabetes do not make enough of another very useful natural hormone called “incretin” eg: GLP 1 [Glucagon like peptide 1, produced by specialized cells of our intestines]. This is where Gliptin medicines are helpful. Gliptins increase the levels of GLP -1 in the blood, by protecting GLP -1 from breaking down. The increased levels of GLP-1, in turn increases insulin production from the islet beta cells [they use the body's own signaling system to boost insulin after meals], decreases glucagon release, decreases gastric emptying and produces satiety, all of which decreases blood glucose levels towards the normal range. This group of drugs does not cause low blood sugar by themselves. Examples of gliptins include: Sitagliptin, Vildagliptin, Saxagliptin, and Linagliptin.
- Sodium-glucose co-transporter 2 (SGLT2) inhibitors: They work by blocking glucose from being reabsorbed by the kidneys. That raises the amount of glucose urinated, and lowers the amount of glucose in the blood. They also reduce body weight and blood pressure to some extent. Examples of SGLT 2 inhibitors include: Canaglifozin, Dapagliflozin, and Empagliflozin.

Insulin Pumps

- When a person has diabetes, it is best to take insulin as close to the way the pancreas delivers it as possible. An insulin pump comes closer to delivering insulin the way a pancreas does, than any other delivery method. Insulin therapy is considered the “gold standard” in diabetes management because it provides the best glucose control. Insulin pumps give smaller and more precise doses throughout the day [basal doses] and higher doses [bolus doses] as required during meals and snacks, and when blood glucose levels are higher than desired.
- An insulin pump is a medical device used for the administration of insulin in the treatment of diabetes mellitus, also known as continuous subcutaneous insulin infusion therapy. The device configuration may vary depending on design. A traditional pump includes: the pump (including controls, processing module, and batteries), a disposable reservoir for insulin (inside the pump), a disposable infusion set, including a cannula for subcutaneous insertion (under the skin) and a tubing system to interface the insulin reservoir to the cannula.
- An insulin pump is an alternative to multiple daily injections of insulin by insulin syringes or an insulin pen and allows for intensive insulin therapy when used in conjunction with blood glucose monitoring and carbohydrate counting.
- **Continuous Glucose Monitoring:** This newer technology provides a precise update of your blood glucose levels every few minutes with a continuous glucose monitor. It is amazing – and can be looked upon as a Glucose Watch. Continuous glucose monitors work with a tiny sensor that a doctor implants in the skin. The sensor sends data about current glucose levels to a wireless device that looks like a pager. The sensors need to be replaced every three to seven days.
- **Artificial Pancreas:** Many companies are working on technology that will fully automate the interaction between a continuous glucose monitor and an insulin pump. You would not have to press the button on your pump when your monitor alerts you. Instead, the monitor would trigger the pump to give the insulin dose directly. The two would operate without your input -- creating what many experts call an "artificial pancreas.”

Diabetes Related Tests

- **Diabetes tests** let you see how the other tools are working. It is important to check the blood sugar and urine sugar from time to time as required to understand your diabetes control.
- **Finger stick blood glucose testing (Self Monitoring of Blood Glucose)** has to be done by patient as per instructions and a record log has to be kept. This is a very useful tool both to patients and physicians in optimizing medicine and insulin dosage.
- **Hemoglobin A1c** is a blood test {done in a standard clinical laboratory} which gives the average blood glucose levels of the past 3 months. This is an important and reliable test which helps us in diagnosis and management of all types of diabetes.
- **Urine ketone testing** by Dipstick method must be done whenever blood glucose is above 250 mg/dl and / or during illness.
- **Urine sugar testing [double void]** has limited use in modern day diabetes
  self care. Consult your doctor for help.
- Continuous glucose monitoring (CGM) is a new tehnological advance which provides your blood glucose readings (levels and trends) through out 24 hours a day for several days.

Your Health A…B...C

A A1C below 7%

A1C (HbA1C) level reflects your blood glucose control over the past 3 months. Consult your doctor to know your A1C goal. Your blood glucose will stay lower when you exercise and eat healthy food. A1C value between 6-7% denotes good control.

B Blood pressure below 130/80

Most diabetics also have high blood pressure. High BP can cause heart attack, stroke, and kidney disease. Consult your doctor if you have high BP.

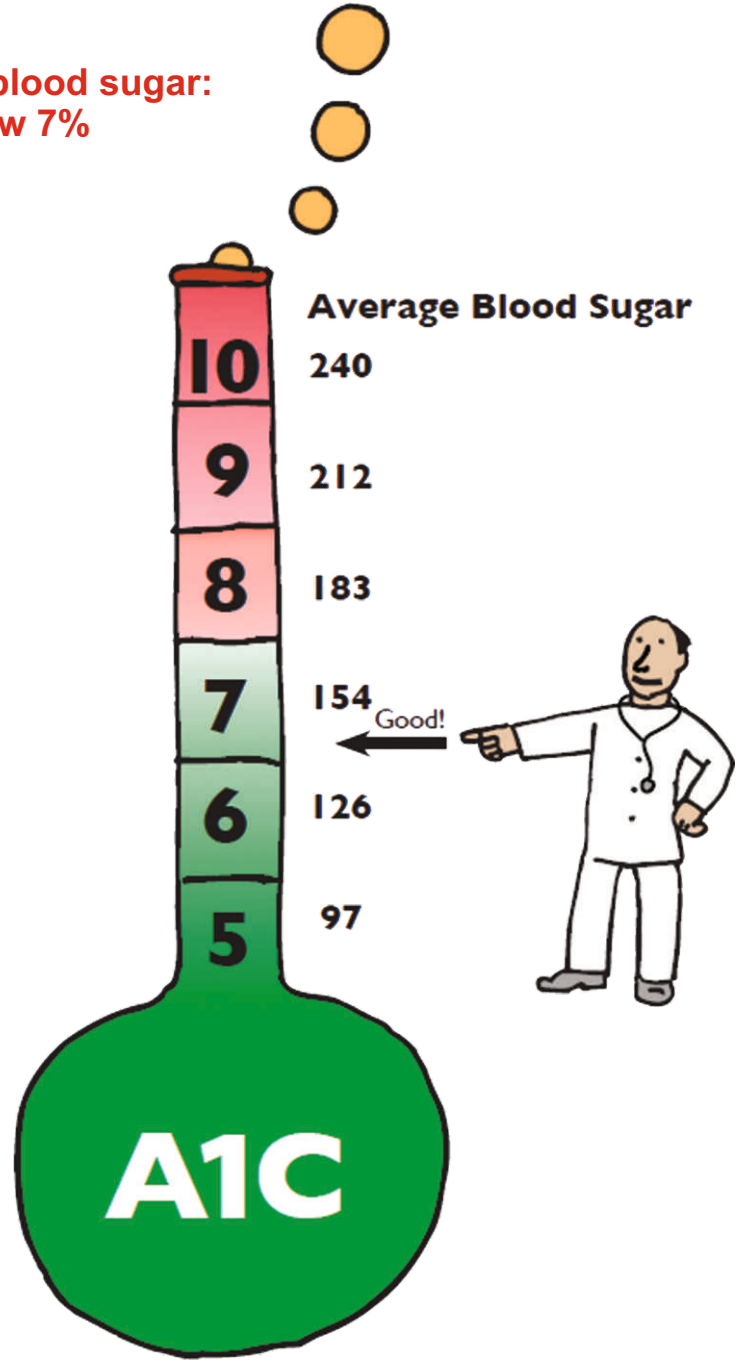
C Cholesterol in control

LDL (bad cholesterol) goal should be below 70.

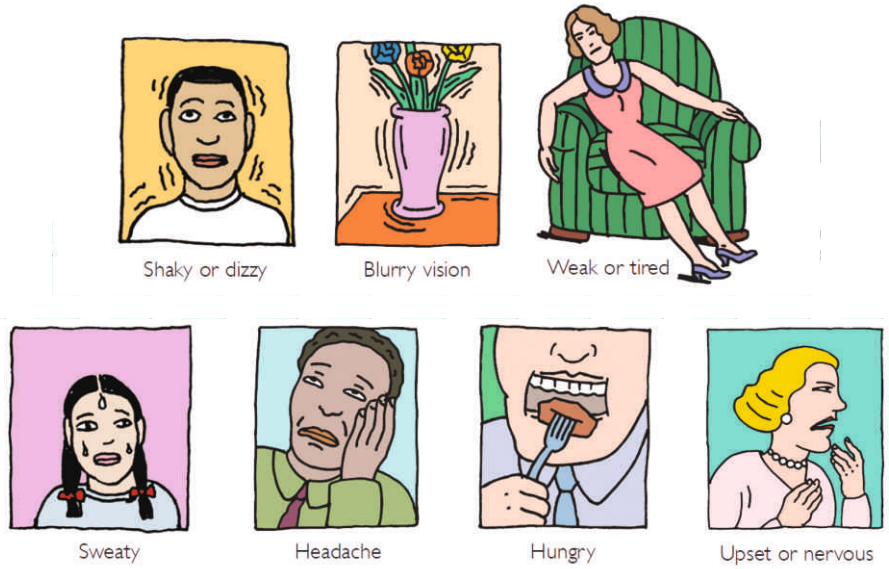
HDL (good cholesterol) goal should be above 50.

Cholesterol abnormalities are common for people with diabetes. Consult your doctor for help.

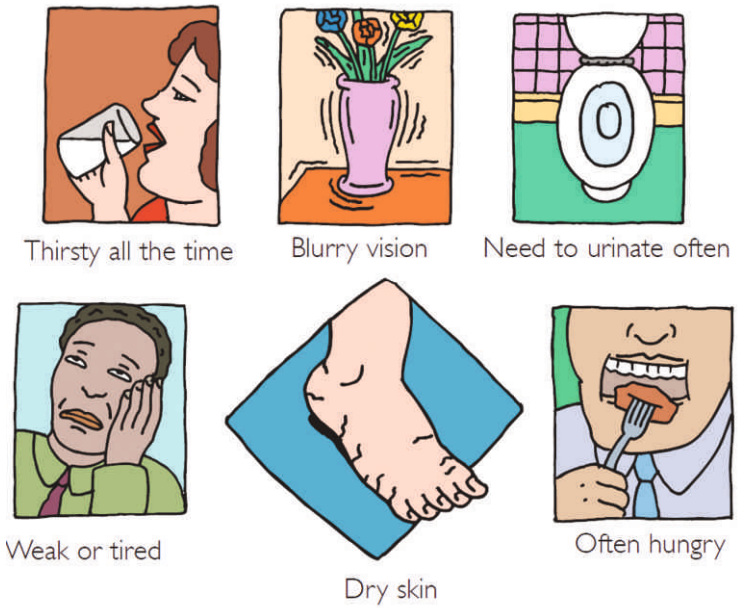
3 months
average blood sugar:
A1C below 7%



Low Blood Glucose



High Blood Glucose



Low Blood Glucose (hypoglycemia)

- When diabetes medicines, food, and exercise get out of balance, the level of glucose in the blood can drop too low. Low blood glucose is the most common day to day problem experienced by people with diabetes. (It can be avoided with a little planning; but if it still occurs, it must be treated).
- **Symptoms:** Shaky, tired or hungry, sweaty, get confused or annoyed for no reason, rapid heart rate. **You could even pass out.**
- If you feel your blood glucose is low, eat some (3-4 teaspoons) of sugar immediately. Other foods you may take are fruit juices, honey, sugar cubes, or a cup of milk with sugar. Once you feel better, eat a snack like biscuits, sandwiches, etc.
- If you are taking insulin and /or other diabetes medicines, please carry on person an *'I am Diabetic'* ID card in your pocket while going out of house / travelling.
- **Glucagon:** If your blood glucose is so low, that you pass out / cannot swallow, you will need glucagon. Glucagon is injected just like insulin and it raises your blood glucose. If you need glucagon, some one else will have to give it to you. Hence your family members, colleagues or friends must be taught how to give 1BGUIBglucagon injections, during low blood sugar emergencies.

High Blood Glucose (hyperglycemia)

- Blood glucose level raises when insulin is less in the body or when insulin is not used properly by your body. The causes for hyperglycemia include: eating excess food, exercising less than usual, stress, or any illness, or taking less than required diabetes medicines
- **Symptoms:** You may feel more thirsty than usual and pass more urine quite often. You may experience loss of weight, blurred vision, susceptibility to some infections, delay in wound healing, tiredness and sleepiness all the time.
- To reduce your high blood glucose, cut down the amount of food you eat. Work with your counselor to make changes in your meal plan. A good exercise program also helps to reduce higher blood glucose levels. If exercise and changes in your diet don't work, your doctor may change the amount of your medication or insulin.

Diabetic Ketoacidosis

► This is a dangerous condition which usually only happens in people with Type 1 diabetes. It can occur when your blood glucose gets very high and is not treated properly. it often happens when you are sick. **If ketoacidosis is not treated, it can lead to coma and even death.** It can be usually avoided by knowing what to do when glucose goes very high.

► Check your blood sugar levels and test your urine for ketones when you: feel more thirsty or hungry than usual, have to pass urine very frequently, lose weight unexpectedly, feel more tired or sleepy than usual, have stomach pain, have vomiting, have smell in your breath, notice your breathing is fast and deep.

Ø **Emergency:** *When your blood sugar is greater than 250 mg/dl , and you have ketones in urine, call / go to your doctor right-away.*

High Blood Pressure (BP)

► Abnormally high blood pressure is referred to as hypertension (HTN). Blood Pressure (BP) rises to a peak called the systolic blood pressure (SBP) and it steadily falls to a lower level called the diastolic blood pressure (DBP) (Eg: SBP / DBP = 120/80).

Classification of blood pressure for adults:

Blood pressure mm Hg.	Systolic	Diastolic
Optimal	<120	<80
Normal	<130	<85
High Normal	130-139	85-89
Hypertension (HTN)		
HTN: Stage 1	140-159	90 - 99
HTN: Stage 2	160 -179	100 - 109
HTN: Stage 3	>180	>110

> Greater than; < Less than

Tips to Control High Blood Pressure (BP)



- **Adverse effects:** Apart from increasing overall morality (deaths), HTN in the general population is associated with increased death rates from heart attacks, heart failure, stroke and renal failures. Treatment of moderate to severe HTN decreases overall mortality, and in particular, the risks of death from stroke, heart failure and kidney failure.
- **Diabetes and hypertension:** HTN appears to be more common in diabetics than in general population, particularly with increasing age. There is a significant increase in HTN in those who develop even early diabetic kidney disease.
- **Treatment:** Treatment of high blood pressure needs to be especially aggressive in diabetes, as these two are a dangerous combination in causing target organ damage (i.e., complications of diabetes). It involves both a change in the life style, and many times, the prescription of drugs.
- **Life style changes:** Bringing about a discipline in terms of diet (including salt restriction); introducing a regular exercise routine into one's everyday living; quitting cigarettes / tobacco in all forms; controlling one's emotions to remain relaxed and peaceful with oneself.
- **Drug therapy:** It offers a choice of several main classes of drugs - ACE inhibitors, calcium antagonists, alpha blockers, beta blockers, diuretics, adrenergic inhibitors, direct vasodilators etc. Your doctor will help you control your blood pressure levels with required prescription of drugs.

Heart Health

- Heart disorders are by-far the most common cause of death throughout the world. They also impair the quality of life of millions of people, restricting activity by causing pain, breathlessness, fatigue, fainting spells, and anxiety.

  Diabetes can cause the cardiac conditions to appear at an early age, worsen the existing condition, or cause more damage and complications than what is seen in general population and non-diabetics, this is known as Diabetic Heart Disease.
- People with diabetes are at higher risk for heart disease, may have additional causes of heart disease, may develop heart disease at a younger age, and may have more severe heart disease
- A wide range of conditions can affect the heart by disrupting its pumping action like, Ischemia-reduced blood supply as a result of Atheroma or plaque formation inside the Coronary arteries, genetic disorders, congenital defects, muscle disorders like cardiomyopathy, injury, infection, tumors, nutritional disorders, poisoning,and drugs etc.)

- Impaired blood supply (Ischemia): The major cause of heart disease in all countries is impaired(reduced) blood supply. The coronary arteries (which supply blood to the heart) become narrowed due to atherosclerosis and parts of the heart muscle are deprived of oxygen. The result of coronary artery disease may be angina pectoris (chest pain on exertion) or, eventually, a myocardial infarction (heart attack).
- Investigation: Heart disease and disorders are investigated by techniques such as auscultation (listening to the heart sounds) and ECG (electrocardiography); stress ECG (tread mill test); heart imaging techniques such as chest X-ray, echocardiography, Doppler colour flow mapping, CT-coronary angiography, coronary angiography, cardiac catheterization; blood-tests; nuclear medicine testing (stress thallium) etc.

Cholesterol and fats

- **Dyslipidemia** is a fat metabolism disorder characterized by abnormal fats or lipids in the blood. This condition is responsible for **coronary heart disease** (CHD) i.e., disease of the arteries supplying blood to the heart itself (coronary = related to heart). Dyslipidemia quite frequently occurs together with diabetes, high blood pressure, cigarette smoking, over-weight and lack of exercises, the other main **"risk factors"** responsible for the development of CHD.
- Increased blood lipids put a burden on the walls of the blood vessels; it also leads to the deposition of lipids on the arterial walls, resulting into a progressive narrowing of the blood vessels (called **"atherosclerosis"**) which reduces the blood flow to the heart and other important parts of the body.
- Blood lipids consists of various fats or fat like substances which serve as suppliers of energy and building material. The important lipids are (1) Cholesterol and (2) Triglycerides. Some cholesterol comes from diet we follow and some is made in our liver.
- **Cholesterol:** It is a fatty substance found in animal fats, whole milk, yolk of the egg, liver and kidney. Cholesterol is also manufactured by your liver. There are two types of cholesterol in human body:
- LDL-Cholesterol (LDL-C) is often refer red to as 'bad' cholesterol, because it deposits cholesterol in the walls of the arteries.
- HDL- Cholesterol (HDL-C) is considered 'good' for your body because it carries cholesterol away from the arteries and to the liver for elimination. A high level of HDL-C has been associated with low incidence of heart attacks.

- **Triglycerides:** These are fatty substances taken in by the body predominantly in animal and vegetable food stuffs, as well as formed in the liver. High level of blood triglycerides ('bad') also increases the vulnerability of a person to heart attacks, particularly in a diabetic.
- **Causes of dyslipidemia:** Diabetes mellitus: uncontrolled, (2) Excessive food intake (food rich in fat, eg. meat, egg yolk, butter, ghee, high intake of sugar), (3) Inheritance from your family, (4) Disorders of the thyroid, kidneys or liver, (5) Lack of physical exercise. (6)Alcohol, (7) Certain medications;a few blood pressure lowering pills, diuretics and hormones (steroids).

Normal values of blood lipids:

Blood lipid mg/dl	Normal or Desirable	Borderline Risk	High Risk
Total Cholesterol	<200	200-239>	240
LDL-Cholesterol	<100	130-159>	160
Triglycerides	<150	150-199>	200
HDL-Cholesterol (M)	>45	35-44	>35
HDL-Cholesterol (F)	>55	45-54	<45

> Greater than; < Less than; M = male; F = female

NOTE : In the presence of coronary heart disease or other disorders related to atherosclerosis, the desirable level of LDL Cholesterol is <70, and desirable level of triglycerides is <150.

- In view of the adverse effects and the common occurrence of dyslipidemia, all diabetics must have their blood lipids measured at least once a year. Once dyslipidemia is recognised both in people with and without diabetes, necessary dietary measures, and later medications need to be instituted to normalise the blood lipids.
- **Diet therapy: Saturated fat** (increases LDL-C) intake is reduced by avoiding red meat (beef, pork) and cutting portion sizes of chicken and fish. Fried foods should be avoided. Vegetable oils that are highly saturated, such as palm oil should be avoided. Low fat dairy products (eg. skimmed milk) can be substituted for whole-milk products. Soft cheese (eg: paneer) should replace butter, solid vegetable fats, cream and high fat cheese.

Cholesterol and fats



- **Polyunsaturated fat** (decreases LDL-C) can be obtained from vegetable oils (eg. sunflower, safflower, soya bean, corn oil) and margarine. Intake of polyunsaturated fat should not exceed 10% of total calories.

- **Monounsaturated fat** (decreases LDL-C) should not exceed 10-15% of total calories. These fats are found in vegetable oils especially olive and groundnut oil.

- **Cholesterol** intake can be reduced by avoiding egg yolks, and organ meats (such as liver, kidney, brain, and sweet bread), trimming all the visible fat from meat and replacing skimmed milk for whole milk.

- Patients with increased blood **triglycerides** usually respond to restrictions of dietary fat and may also require decreased intake of simple sugars and alcohol.

  Fresh fruits, vegetables and whole grain products should be used to increase variety and provide nutrients and fibre.

- **Weight loss** is beneficial if the patient is over weight. Total calories should be adjusted for gradual weight loss. Elevated triglycerides often respond well. HDL-C may increase with weight loss.

- **Exercise** is often helpful in lowering triglycerides and LDL-C levels, and in raising HDL-C. It can also contribute to weight loss.

- **Good diabetes control** with adequate meal planning, physical activity, diabetes pills and/or insulin is critical factor towards normalising your blood lipids levels.

- **Discontinuation of smoking** raises HDL-C levels, besides its numerous other health benefits.

- **Specific drug therapy** is considered, if maximal diet, weight reduction, exercise and other efforts mentioned above do not reduce serum lipids to goal levels after an adequate trial (of 3 - 6 months). Examples of such drugs include; Resins (Cholestyramine, Colestipol), Nicotinic acid, HMG CoA reductase inhibitors or statins (Lovastatin, Pravastatin, Simvastatin, Atorvastatin, Rosuvostolin), Fenofibrate.

Smoking

- **Smoking** Is Believed To Be The Chief, Single Avoidable Cause Of Death In Our Society. Of Every 1000 Young Men Who Smoke, One Will Be Murdered, Six Will Die On The Roads, But 250 Will Be Killed Before Their Time By Tobacco.

  Remember: Each Cigarette Reduces Your Life By 5.5 Minutes.

- **Heart:** Cigarette Smoking Has Invariably Been Found To Be A Major Risk Factor In The Causation Of Heart Attack, Stroke And Diseases Of Blood Vessels.

- **Sudden Death:** Is The Most Frequent Clinical Event Associated With Smoking. Smoking Makes A Person Vulnerable To Cancer (mostly Of The Lungs), Chronic Bronchitis, And Various Diseases Of The Heart. The Risk Of Cancer And Atherosclerosis Is About 20 Times Greater In Smokers Than In Non-smokers.

- **Diabetes And Smoking:** Smoking And Diabetes Are A Dangerous Combination, Especially If You Also Have Hypertension. In Diabetes, Smoking Worsens The Occurrence And Severity Of Virtually All Complications (heart Attacks, Strokes, Gangrene And Amputations, Kidney Disease, Eye Disease Etc.) And Significantly Increases The Chance Of Premature Death. In Fact, Stopping Smoking Can Protect You More From Disease And Death, Than By Tightening Blood Glucose Control Or Treating Hypertension.

Stop Smoking.

Alcohol

- **Alcohol Is Poison.** Nearly 40%-50% Of People Consuming Alcohol Suffer From Temporary Alcohol Induced Problems. 10% Of Drinkers Are Known To Have Persistent Alcohol Induced Problems. Risk Of Cancer Of Various Organs Like Esophagus, Stomach, Liver, Pancreas, Breast And Neck Increases By 10 Times Compared To Non-drinkers.

- Alcohol Is Known To Affect The Heart And May Lead To Condition Called Cardiomyopathy. Alcohol Intake Adversely Affects Blood Fat Levels. It Increases Triglyceride Levels And Decreases The Level Of Hdl (good) Cholesterol. This Is Especially Bad In People With Diabetes, Since They Are Already Prone To Adverse Changes In Their Blood Fat Levels **(dyslipidema).**

- **Alcohol And Diabetes:** In Addition To The Above Mentioned Multiple Harmful Effects Of Alcohol, It Is Important To Be Aware Of Its Effects On Management Of Diabetes. Alcohol Has No Nutritional Value. It Provides No Carbohydrates, Proteins Or Fat. Alcohol Gives 7 Empty Calories For Every Gram Consumed And It Would Not Leave Enough Calories For Eating Other Foods Necessary For A Nutritionally Balanced Diet. Alcohol Blocks The Ability Of Liver To Release The Reserve Stores Of Glucose Into The Blood Stream. This Can Result In Hypoglycemia Or Low Glucose Reaction. Some Alcoholic Beverages Such As Beer, Sweet Wines, Liquors And Cordials Contain Large Amounts Of Concentrated Sugar And Their Consumption Will Cause Very High Levels Of Blood Glucose.

Avoid Alcohol.

Obesity

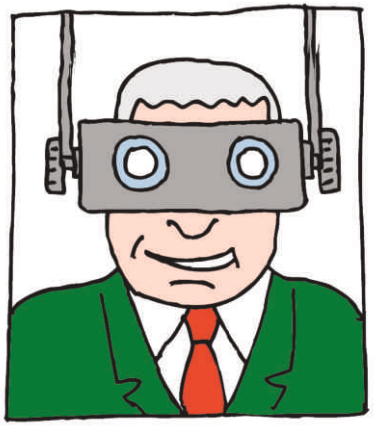
- **Obesity** is an excess amount of body fat. If a person weighs more than 20% of his ideal body weight, he or she is considered obese.
- **Causes of obesity:** Genetic, endocrine, neurologic, psychological, and environmental factors contribute to obesity **(overeating and physical inactivity).**
- **Problems related to obesity:** Obesity progressively increases the risk of death in all people. It is associated with an increased incidence of coronary heart disease, congestive heart failure, stroke, diabetes, hypertension, dyslipidemia, arthritis, gout and gall stones and some cancers. Obesity by itself is a risk factor for the above mentioned ailments, independent of other risk factors like smoking, cholesterol level, age etc.
- **Diabetes and obesity:** Obesity is an important triggering factor for developing diabetes in people who have inherited (bad genes) a tendency for diabetes. Thus it has been said that apparently most adult onset diabetes is preventable and probably a majority is curable, provided that complete and permanent reversal of obesity is achieved. Obesity impairs action or effectiveness of the insulin produced in the body or taken as insulin injections.
- Obesity can be managed by healthy meal planning, regular physical activity and behaviour modification. The goal is to achieve and then maintain the **ideal body weight.**

Eye Care

- Eye problems can occur due to diabetes. Uncontrolled diabetes and high blood pressure can damage the small blood vessels in your retina, the back part of your eye. This is called diabetic retinopathy.

  Have a complete annual dilated eye exam every year. The eye examination should not be confused with the eye glass (spectacles) check. The eye exam should include tests to check for signs of cataract and glaucoma and fundoscopy. Consult your doctor for further details and referrals to eye hospitals.
- With respect to your eye care, keep the following in mind:
  - Get your eyes examined regularly (once a year).
  - Fundoscopic exam and fundus photography
  - Examination to check for cataract.
  - Examination of intra ocular tension is a must to prevent glaucoma.
- Do not ignore early signs of infections or eye irritation.
- Use protective glasses to avoid dust and infection.
- Consult your doctor as soon as you feel blurring or pain in eye- balls.

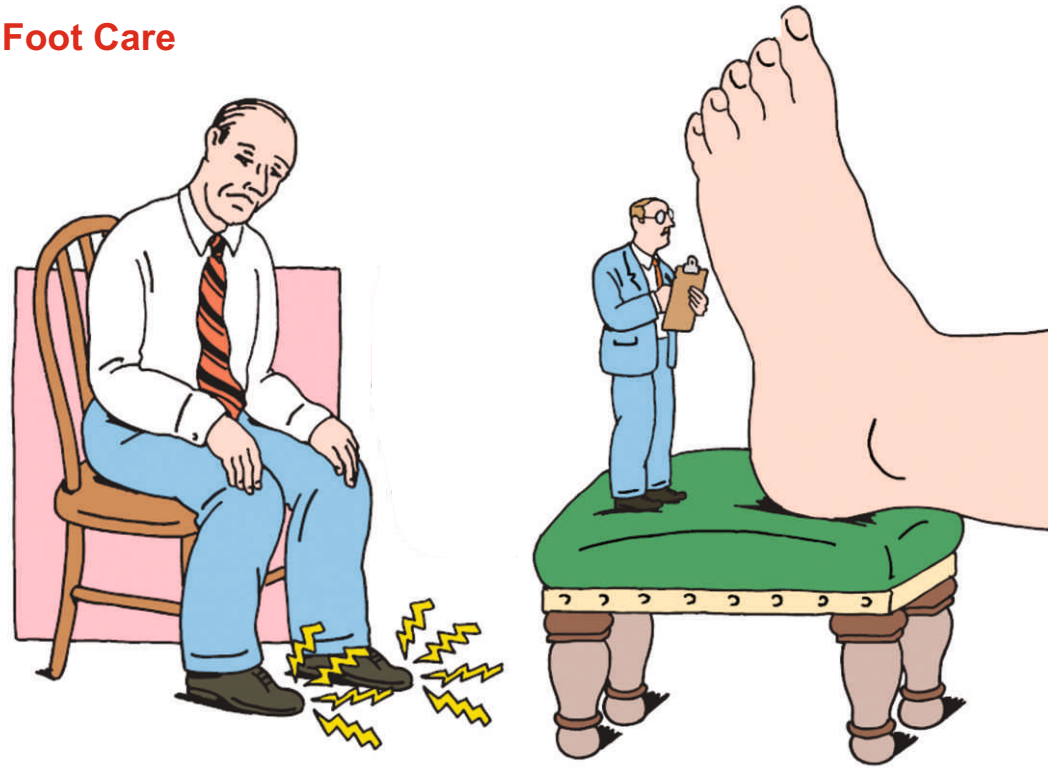
Eye Care



Dental Care



Foot Care



Dental Care

► Teeth and dentures can cause special problems in people with diabetes, if they are not taken proper care. Periodontal disease is associated with poor diabetes control, insulin resistance and cardiovascular disease. Periodontal abscess can lead to alveolar bone loss, and delayed healing. Caries or infected teeth can be a source of infections in other organs and cause pneumonia, sinusitis, enteritis, meningitis etc.

► With respect to your dental care, keep the following in mind:

- Get your teeth / dentures examined regularly (once a year).
- Get comprehensive periodontal examination.
- Keep your teeth always clean and avoid infection.

Foot Care

► People with diabetes can develop many different foot problems. Most often,
the cause for foot problems is nerve damage and poor blood flow. This is compounded by various infections of the feet. Due to your gait, mobility, advancing age and weight of your body, changes in the shape of your feet or toes can also occur. Even ordinary problems can lead to serious complications.

► Feet should be examined regularly for loss of protective sensation, to identify risk factors predictive of ulcers and amputation.

► With respect to your foot care, keep the following in mind:

- Examine your foot daily and regularly.
- Use a towel to dry your legs after washing.
- Give attention to nail and foot hygiene.
- File your toe nails straight across.
- Wear good fitting shoes when you go outdoors.
- Take prompt treatment of infections.
- Do not ignore early signs of infections or skin discoloration.
- Consult your doctor as soon as you feel numbness or prickly sensation in your feet.

DIABETES CHALLENGE

► Initially there is nothing easy about managing your diabetes. It takes time, effort and courage. But you can do it with the help of your medical team and family.



Thank you - Respects - Good Health

Diabetes Day Care Unit

► **Diabetes Day Care Unit** currently provides two special health care programs targeted to help people with diabetes and their families to obtain the best in comprehensive and advanced diabetes health care. These programs are administered by a 'team' of highly experienced diabetologists, dieticians and
nurse educators supported by ultra- modern laboratory services and super specialty consultant referrals.

Diabetes annual health review

► **Program:** During the course of single day, complete physical examination and investigations for diabetes, blood glucose control and associated problems (heart, kidneys, eyes, feet etc.), evaluation and counselling by dieticians and diabetes educators, and review and treatment guidance by expert diabetologists are provided.

► **Participation:** Beneficial to all people with diabetes once a year ('Annual Review' and 'Follow up')

► **Dates and Time:** Weekdays (Mon to Sat)

Diabetes self care basic and intensive

► **Program:** Basic and intensive diabetes health education and self care training program for the patients and family, including lectures, demonstrations, discussions and guidance on various practical aspects of health care; the nature of diabetes, its proper management in terms of meal planning, physical activity, insulin injections and oral drugs, newer diabetes medicines, insulin pumps and continuous glucose monitoring special situations and emergencies (hypoglycemia, ketoacidosis, sick day guidelines), self monitoring of diabetes (urine and blood glucose testing), prevention of complications (heart, kidneys, eyes, feet, infections.....etc.); supported by easy to understand and follow learning source materials, booklets and video.

► **Participation:** Beneficial to all 'new' diabetes patients and those eager to learn more about self management of diabetes ('Initial review' and 'Enrichment').

► **Dates and Time:** Please enquire

Samatvam Endocrinology Diabetes Center
Jnana Sanjeevini Medical Center And Diabetes Hospital

Endocrinology Metabolism Unit

- Human problems that will be addressed to provide medical and health care of highest possible quality, include:
- **Dibetes Mellitus:** Initial visit (IDDM, NIDDM, gestational diabetes, Secondary diabetes); Follow up care (for both day to day management and long term prevention / management of chronic complications of diabetes, cardiovascular disease, peripheral vascular disease, cerebrovascular disease, nephropathy, retinopathy, neurophaty, diabetic foot).
- **Thyroid Disorders:** Hyperthyroidism, hypothyroidism, thyroiditis, thyroid neoplasia.
- **Reproductive Endocrinology:** Male and female infertility, hypogonadism, impotence, menstrual disorders, menopausal disorders, hirsuitism, gynaecomastia, ambiguous genitalia / intersex.
- **Paediatric Endocrinology :** Growth and development, short stature, disorders of puberty, endocrine metabolic disorders of infancy childhood and adolescence.
- **Adrenal Disorders:** Addisons disease, Cushings syndrome, hyperaldosteronism, pheochromocytoma, congenital adrenal hyperplasia.
- **Pituitary and neuroendocrine disorders :** Pituitary and hypothalamic tumors, hypopituitarism, diabetes insipidus, SIADH
- **Obesity:** and associated metabolic and medical problems
- **Dyslipidemia:** Atherosclerosis and vascular diseases
- **Bone and mineral metabolism:** Hyperparathyroidism, hypoparathyroidism, hypercalcemia, hypocalemia, rickets na dosteomalacia, renal osteodystrophy, flurosis, kidney stones.
- **Others:** Hypoglycenia and pancreatic endocrine tumors; Endocrine hypertension; Inborn errors of metabolism and other metabolic disorders.

Notes:

Notes: